Benjamin Boe

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Education

Ph.D., Astronomy, Institute for Astronomy, University of Hawai'i (IfA/UH)	August 2020
Advisor: Shadia Habbal	
Thesis: "Solar Wind Formation in the Corona"	
M.S., Astronomy, IfA/UH	May 2018
B.S., Physics and Mathematics, University of Puget Sound (UPS)	May 2015
Cum laude and Honors in Physics	

Professional Appointments

Assistant Professor, Wentworth Institute of Technology (WIT)	August 2023—present
NSF Postdoctoral Fellow, IfA/UH	October 2020—March 2023
Faculty Lecturer, Physics & Astronomy, UH	August 2021—December 2021

Grants

Collaborative PI, NSF SHINE Grant (AGS 2602403)	$80,961	2026–2029
Additional $519,030 in lead Collaborative Research grant to Shadia Habbal (AGS 2602402)		
Lead PI, NSF SHINE Grant (AGS 2501212)	$385,959	2025–2028
Additional $182,730 in Collaborative Research grant to Cooper Downs (AGS 2501213)		
PI, NSF AGS Postdoctoral Research Fellowship (AGS-PRF 2028173)	$190,000	2020–2023
Student, NSO Graduate Fellowship (AURA grant N97991C)	$122,099	2018–2020
Student, NSF Research Experience for Undergraduates (REU)	$5,000	2014
Student, Washington State NASA Space Grant	$2,250	2013

Publications

Peer-Reviewed Articles

Solar Physics

1. **Boe, B.**, Habbal, S., Druckmüller, Štarha, P., Štarha, M., Hoderová, Constantinou, S., Ayars, E., Casillas, C., *The 2023 Australian Total Solar Eclipse: Line Emission of Fe XIV, Fe X and Fe XI out to 6 solar radii,* ApJ, 997, 171, 2026. 10.3847/1538-4357/ae2658

2. Benavitz, L., **Boe, B.**, Habbal, S. R. *Total Solar Eclipse White-light Images as a Benchmark for Potential Field Source Surface Coronal Magnetic Field Models: An In-depth Analysis over a Solar Cycle.* ApJ, 974, 178, 2024. 10.3847/1538-4357/ad71c6

3. Zhu, Y., Habbal, S., Ding, A., Yamashiro, B., Landi, E., **Boe, B.**, Constantinou, S., and Nassir, M., *Spectroscopic Observations of the Solar Corona during the 2017 August 21 Total Solar Eclipse: Comparison of Spectral Line Widths and Doppler Shifts between Open and Closed Magnetic Structures,* ApJ, 699, 122, 2024. 10.3847/1538-4357/ad3424

4. **Boe, B.**, Downs, C., Habbal, S., The Solar Minimum Eclipse of 2019 July 2. III. *Inferring the Coronal T_e with a Radiative Differential Emission Measure Inversion,* ApJ, 951, 55, 2023. 10.3847/1538-4357/acd10b

5. Cooper, J., Habbal, S., **Boe, B.**, Angelopoulos, V., Sibeck, D., Paschalidis, N., Sittler, E., Jian, L., and Killen, R., *Lunar Solar Occultation Explorer (LunaSOX)*, Frontiers in Astronomy and Space Sciences, 10, 1163517, 2023, 10.3389/fspas.2023.1163517

6. **Boe, B.**, Habbal, S., Downs, C., Druckmüller, M., *The Solar Minimum Eclipse of 2019 July 2. II. The First Absolute Brightness Measurements and MHD Model Predictions of Fe X, XI, and XIV out to 3.4 $R_\odot$*, ApJ, 935, 173, 2022. 10.3847/1538-4357/ac8101

7. **Boe, B.**, Yamashiro, B., Druckmüller, M., and Habbal, S., *The Double-bubble Coronal Mass Ejection of the 2020 December 14 Total Solar Eclipse*, ApJ Letters, 914, L39, 2021. 10.3847/2041-8213/ac05ca

8. Habbal, S., Druckmüller, M., Alzate, N., Ding, A., Johnson, J., Štarha, P., Hoderová, J., **Boe, B.,** Constantinou, S., and Arndt, M., *Identifying the Coronal Source Regions of Solar Wind Streams from Total Solar Eclipse Observations and in situ Measurements Extending Over a Solar Cycle*, ApJ Letters, 911, L4, 2021. 10.3847/2041-8213/abe775

9. **Boe, B.**, Habbal, S., Downs, C., Druckmüller, M., *The Color and Brightness of the F-Corona Inferred from the 2019 July 2 Total Solar Eclipse*, ApJ, 912, 44, 2021. 10.3847/1538-4357/abea79

10. **Boe, B.**, Habbal, S., Druckmüller, M., *Coronal Magnetic Field Topology From Total Solar Eclipse Observations*, ApJ, 895, 123, 2020. 10.3847/1538-4357/ab8ae6

11. **Boe, B.**, Habbal, S., Druckmüller, M., Ding, A., Hodérova, J., and Štarha, P., *CME-induced Thermodynamic Changes in the Corona as Inferred from Fe XI and Fe XIV Emission Observations during the 2017 August 21 Total Solar Eclipse*, ApJ, 888, 100, 2020. 10.3847/1538-4357/ab5e34

12. **Boe, B.**, Habbal, S., Druckmüller, M., Landi, E. , Kourkchi, E., Ding, A., Štarha, P., & Hutton, J., *The First Empirical Determination of the Fe^{10+} and Fe^{13+} Freeze-in Distances in the Solar Corona*, ApJ, 859,155, 2018. 10.3847/1538-4357/aabfb7

Long Period Comets

1. **Boe, B.**, Jedicke, R., Meech, K., Wiegert, P., Weryk, R., et al., *The orbit and size-frequency distribution of long period comets observed by Pan-STARRS1*, Icarus, 333, 252, 2019. 10.1016/j.icarus.2019.05.034

Galaxy Evolution

1. Shields, D., **Boe, B.**, Pfountz, C., Davis, B., et al., *Spirality: A novel way to measure spiral arm pitch angle*, Galaxies, 10, 100, 2022. 10.3390/galaxies10050100

Musical Acoustics

1. **Boe, B.**, and Randy Worland, *Experimental and numerical analysis of the effect of length in musical drumhead coupling*, Proceedings of Meetings on Acoustics, 20, 1, 2014. 10.1121/1.4884779

Non-Peer-Reviewed

White Papers for the 2024 Solar and Space Physics (Heliophysics) Decadal Survey

1. **Boe, B.**, Habbal, S., and Ding, A., *The Need for a New Generation of Space-based Visible and Near-IR Emission Line Observations of the Corona,* 2023, 10.3847/25c2cfeb.fb89d0ac

2. Habbal, S., **Boe, B.**, Haggerty, C., and Ding, A., *A Total Solar Eclipse Earth-Based Mission: Multi-wavelength Observations from Land, Sea and Air to Probe the Critical middle Corona,* 2023, 10.3847/25c2cfeb.70c8c8e3

3. Cooper, J., Habbal, S., **Boe, B.,** et al., *Heliophysics from, by, and of the Moon in the Age of Artemis: Lunar Solar Occultation Explorer (LunaSOX),* 2023, 10.3847/25c2cfeb.beac35c4

Talks, Presentations, and Workshops

Workshops

1. Invited Workshop Participation, “Eclipses and Beyond: Unveiling the Mysteries of the Sun's Visible Corona”, organized and sponsored by the International Space Science Institute (ISSI), Bern, Switzerland, 2025.

Colloquia

1. “Coronal science with Narrowband and White-light imaging from Eclipses”, Invited Colloquium. University of Massachusetts Lowell, virtual, 2025
2. “The Scientific Value of Total Solar Eclipses”, Invited Colloquium. California State University, virtual, 2022
3. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Invited Colloquium. University of Puget Sound, Tacoma, WA, 2020
3. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Invited Colloquium. Bridgewater State University, virtual, USA, 2020

4. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Invited Colloquium. Aberystwyth University, Aberystwyth, Wales, 2019
5. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Colloquium. Institut d'Astrophysique Spatiale*,* Orsay, France, 2019
6. “Total Solar Eclipses as a Tool to Study Coronal Physics”, Colloquium. Brno University of Technology*,* Brno, Czech Republic, 2019
7. “Studying Solar Coronal Physics with Total Solar Eclipses”, Invited Colloquium. University of Puget Sound, Tacoma, WA, 2019
8. “Testing Solar System Formation With Manx Comets”, Colloquium. UH NASA Astrobiology Institute Seminar, University of Hawai'i, 2016

Talks and Posters

1. “The Electron Temperature Inferred with Fe X, Fe XI and Fe XIV Emission Throughout the Middle Corona From the 2023 Australian Total Solar Eclipse”, Talk. American Geophysical Union (AGU) Meeting, New Orleans, LA, 2025
2. “Coronal science with Narrowband and White-light imaging from Eclipses”, Invited Talk. Center for Astrophysics | Harvard & Smithsonian (CfA), Cambridge, MA, 2025
3. “Advances in Total Solar Eclipse Narrowband Observations: Introducing a Radiative Differential Emission Measure”, Talk. AGU Meeting, District of Columbia, 2024
3. “Insights from a Total Solar Eclipse Research Expedition: Educating Students and Engaging the Public”, Talk. American Association of Physics Teachers (AAPT) meeting, Boston, MA, 2024
4. “The First Absolute Brightness Measurements and MHD Model Predictions of Fe X, XI, and XIV out to 3.4 Rs”, Talk. Triennial Earth-Sun Summit, Bellevue, WA, 2022
5. “The First Absolute Brightness Measurements and MHD Model Predictions of Fe X, XI, and XIV out to 3.4 Rs”, Poster. Solar Heliospheric and Interplanetary Environment (SHINE) Workshop, Honolulu, HI, 2022
6. “The Double-bubble Coronal Mass Ejection of the 2020 December 14 Total Solar Eclipse”, Poster. AGU Meeting, hybrid (virtual), 2021
7. “The Double-bubble Coronal Mass Ejection of the 2020 December 14 Total Solar Eclipse”, Poster. SHINE Workshop, virtual, 2021
8. “Coronal Magnetic Field Topology From Total Solar Eclipse Observations”, iPoster. American Astronomical Society (AAS) Solar Physics Division Meeting, virtual, 2020
9. “Thermodynamic Changes in the Corona during the 2017 August 21 Total Solar Eclipse ”, Talk. AAS Meeting, Honolulu, HI, 2020
10. “CME Induced Thermodynamic Changes in the Corona as Inferred from Fe XI and Fe XIV Emission Observations during the 2017 August 21 Total Solar Eclipse”, Poster. AGU Meeting, San Francisco, CA, 2019
11. “The Orbit and Size-Frequency Distribution of Long Period Comets Observed by PS1”, Talk. EPSC-DPS Joint Meeting, Geneva, Switzerland, 2019

12. “CME Induced Thermodynamic Changes in the Corona as Inferred by Fe XI and Fe XIV Emission Observations from the 2017 August 21 Total Solar Eclipse”, Poster. SHINE Workshop, Boulder, CO, 2019
13. “First Empirical Determination of the Fe 10+ and Fe 13+ Freeze-in Distances in the Solar Corona”, Poster. SHINE Workshop, Cocoa Beach, FL, 2018
14. “First Empirical Determination of the Fe 10+ and Fe 13+ Freeze-in Distances in the Solar Corona”, Talk. Asia Oceania Geosciences Society Meeting, Honolulu, HI, 2018
15. “Breaking the Wall of Solar Wind Formation”, Talk. Falling Walls Lab, UH, 2017
16. “Testing Solar System Formation Models Using Pan-STARRS1 Detections of Nearly Inactive Manx Comets”, Talk. Division of Planetary Sciences Meeting, Pasadena, CA, 2016
17. “Pitch Angle Survey of GOODS Spiral Galaxies”, Poster. AAS Meeting, Seattle, WA, 2015
18. “Pitch Angle Survey of GOODS Spiral Galaxies”, Poster. Murdock College Science Research Program (MCSRP) Conference, Vancouver, WA, 2014
19. “Experimental and numerical analysis of the effect of length in musical drumhead coupling”, Poster. Acoustical Society of America Meeting, San Francisco, CA, 2013
20. “Coupled Vibrations Between Musical Drumheads”, Talk. MCSRP Conference, Vancouver, WA, 2013

Observing Experience and Telescope Time

Daniel K Inouye Solar Telescope (DKIST)

Three successful proposals as PI (Cycles 2, 3 and 4) and one as Co-I (Cycle 4)

Total Solar Eclipse Expeditions

Preparation in lab and on-site operation, alignment, and calibration of instrumentation

Kerrville, Texas	April 8th, 2024
Exmouth, Australia	April 20th, 2023
Antarctic Ocean	December 4th, 2021
Vicuña, Chile	July 2nd, 2019
Alliance, Nebraska	August 21st, 2017
Halmahera, Indonesia	March 9th, 2016

Maunakea Observing

Subaru HSC, summit observing	January 3rd and 4th, 2016

Student Projects

Advised 8 undergraduate students in academic research

- 3 are now in graduate school
- One published in ApJ (Benavitz et al. 2024, see above)
- 2 were senior projects, 1 internally funded summer project, 1 NSF REU, and 3 Co-ops (semester long internship)

Advised one graduate student

- Student of my former PhD advisor (Shadia Habbal)
- Provided DKIST data for their thesis, and meet with them regularly

Teaching Experience

Physics Courses

Modern Physics (PHYS 3100) — 1 section at WIT
Special relativity, blackbody radiation, atomic structure, photoelectric effect, scattering processes, introductory quantum mechanics.

Engineering Physics 1 (PHYS 1250) — 8 sections at WIT with labs
Mechanics: kinematics, Newton's laws, work–energy, momentum, rotational dynamics.

Engineering Physics 2 (PHYS 1750) — 3 sections at WIT with labs
Electric and magnetic fields, DC circuits, and geometric optics

Astronomy Courses

Foundations of Astrophysics I: The Solar System (ASTR 214) — 1 section at UH
Calculus-based introduction to orbital mechanics, tidal interactions, planetary interiors and atmospheres, small bodies, hydrostatic/thermal equilibrium, and solar & heliospheric physics.

Introduction to Astronomy (ASTR 110; PHYS 2000) — 3 UH lecture; 2 WIT lecture+lab
Survey of sky motion, astronomical methods and instrumentation, planetary and stellar physics, galaxies, and cosmology

Introduction to Astronomy Lab (ASTR 110L) — 3 sections at UH
Hands-on observational laboratory including nighttime telescope operation and data-driven activities.

Space Exploration (PHYS 2300) — 1 section at WIT with lab
Original course integrating algebra-based physics with spaceflight history, emphasizing spacecraft dynamics and mission design.

Service

WIT Honors Program Council member	starting Fall 2026
Course Coordinator for Physics 1250 and 1750, 3 semesters	2025–present
WIT Liaison, Massachusetts Space Grant Consortium (NASA)	2024–present
WIT Physics Curriculum Committee member	2023–present
Astronomy Club Faculty Advisor, Colleges of the Fenway	2023–2025

Referee

- The Astrophysical Journal, 11 articles
- Solar Physics, 5 articles
- NSF Grant Proposals, 2 panels, 3 ad hoc
- DKIST Science Review, 1 panel

Selected Press and Outreach

Articles:
- The Guardian article about 2023 eclipse expedition, link
- NASA article about Habbal et al. (2021), with interview, link
- Forbes article about Boe et al. (2020b), link

Podcasts and Interviews:
- Wentworth Research Podcast, June 1st, 2026, interview, link
- NASA's Curious Universe Podcast, May 3rd, 2024, interview, link
- Texas Standard (public radio), live on-air interview, April 8th, 2024, link
- WABI Bangor Maine Local News, interview, March 22nd, 2024, link

Outreach and Talks:
- Astronomy club, with my mentorship, hosted eclipse viewing event on WIT campus, April 8th, 2024
- Kerrville, Texas, public talk at eclipse festival, April 7th 2024
- Boston Museum of Science, public talk, March 26th, 2024

Honors, Awards, and Scholarships

Sigma Pi Sigma Physics Honor Society		2015
University of Puget Sound Dean's list		2014
Raymond & Olive Seward Scholarship	$3,500	2014
Acoustical Society of America poster award	$500	2013
John Van Zytveld Award in Physical Science	$2,000	2013
Thomas & Hilda Jack Scholarship	$2,750	2013
Puget Sound President's Scholarship	$64,000	2011–2015
Kilworth Foundation Scholarship	$2,500	2011